

Ambiguous Healthcare EOB Test Document

Purpose

This synthetic document is designed to test a cascading PII/PHI detection pipeline (Regex → Presidio → MedSpaCy → GLiNER → Ollama/Qwen). It intentionally contains ambiguous entities that require contextual understanding.

Document

Explanation of Benefits (EOB)

Patient Name: Michael Jordan
DOB: 1967-04-15
SSN: 881-32-3610
Member ID: MCA-78291-XX
Visit Date: 2026-07-28
Primary Physician: Dr. Brown
Consulting Specialist: Dr. Green
Provider: Apollo
Laboratory: ABC Diagnostics
Diagnosis: Stage II
Procedure: Revision
Medication: NovoRapid
Insurance Provider: United
Policy Number: UH-8827-191A
Billing Contact: Sarah
Emergency Contact: Jordan
Phone: +1 (555) 742-8821
Email: sarah.billing@example.com
Address: 418 Lake View Drive, Austin, TX 78701

Clinical Notes

The patient was transferred from Apollo after evaluation by Dr. Green. Brown recommended continuing NovoRapid while reviewing Revision results. Stage II remains stable. ABC Diagnostics will forward reports next week. United approved the procedure yesterday. Sarah confirmed insurance authorization. Jordan confirmed the patient's previous prescription.

Amount Billed: \$2,845.00
Amount Covered: \$1,935.00
Patient Responsibility: \$910.00

Expected Detector Behavior

Regex: DOB, SSN, Policy Number, Phone, Email.
Presidio: Michael Jordan, Sarah, Address.
MedSpaCy: NovoRapid, Stage II (if supported).
GLiNER: Apollo, ABC Diagnostics, United (if recognized).
LLM/Ollama: Brown, Green, Jordan, Revision, Apollo/United if unresolved after previous detectors.

What to Verify

1. High-confidence entities are masked.
2. Low-confidence entities continue to the next detector.

3. Only unresolved entities reach the LLM.
4. Context windows help the LLM resolve ambiguous entities.
5. Field labels such as 'Amount Billed' should not become unresolved semantic entities unless intentionally designed.